

Math 133 Written Homework 4

Week 6: Limits and Continuity

Due Thursday, October 8, 2026, at 11:59 p.m.

Show your work, including algebraic simplifications and reasons for continuity conclusions. List collaborators, time spent, and any use of generative AI with your submission. Your submitted work must be your own. Late work will not be accepted.

For limits, give an exact value, $+\infty$, or $-\infty$ when appropriate. If the one-sided limits disagree, state that the two-sided limit does not exist. Identify any indeterminate form encountered; an indeterminate form is not a final answer.

When evaluating limits, justify any use of continuity or limit laws by explaining why the relevant conditions are satisfied.

1. Evaluate each limit.

(a) $\lim_{x \rightarrow -2} \frac{3x^2 + x - 4}{x^3 + 10}.$

(b) $\lim_{x \rightarrow 5} (2x - 3)\sqrt{x + 4}.$

2. Evaluate each limit by factoring and simplifying.

(a) $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}.$

(b) $\lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{x^2 - 4}.$

3. Evaluate each limit.

(a) $\lim_{x \rightarrow 7} \frac{\sqrt{x + 2} - 3}{x - 7}.$

(b) $\lim_{x \rightarrow 2} \frac{\frac{1}{x+3} - \frac{1}{5}}{x - 2}.$

4. Evaluate each limit.

(a) $\lim_{x \rightarrow 2^+} \left(\frac{1}{x - 2} - \frac{4}{x^2 - 4} \right).$

(b) $\lim_{x \rightarrow 3^+} (x - 3) \frac{x + 2}{x^2 - 9}.$

5. Evaluate

$$\lim_{x \rightarrow 1} \left[\frac{1}{x-1} + \left(\frac{x+1}{x-1} \right)^2 \right].$$

6. State the domain of each function and determine where it is continuous:

$$f(x) = \frac{\sqrt{x+4}}{x-1}, \quad g(x) = \frac{\ln(x-2)}{x+1}.$$

Evaluate $\lim_{x \rightarrow 0} f(x)$ and $\lim_{x \rightarrow 3} g(x)$.

7. Let

$$h(x) = \frac{|x+2|}{x+2}.$$

State the domain and determine whether h is continuous at $x = -2$. If it is discontinuous, classify the discontinuity as removable or a jump, and state which conditions of continuity fail.

8. Consider the functions

$$A(x) = \begin{cases} x^2 - 2, & x < 3, \\ 5, & x = 3, \\ 2x + 1, & x > 3, \end{cases} \quad B(x) = \begin{cases} 2x - 1, & x \leq 3, \\ x + 4, & x > 3. \end{cases}$$

For *each* function state the domain, and find the function value at 3. Determine if the function is continuous at $x = 3$. If it is discontinuous, classify the discontinuity as removable or a jump, and state the condition of continuity that fails.

9. Let $p(x) = x^3 + 2x - 6$. State its domain and explain why it is continuous on $[1, 2]$. Use the Intermediate Value Theorem to show that $p(c) = 0$ for at least one $c \in (1, 2)$.

10. Find all real values of k for which

$$H_k(x) = \begin{cases} kx^2 + 1, & x \leq 2, \\ 2x + 3, & x > 2 \end{cases}$$

is continuous on all of $\mathbb{R}$. For your value of k , verify all three continuity conditions at 2 and explain why there are no other discontinuities.